

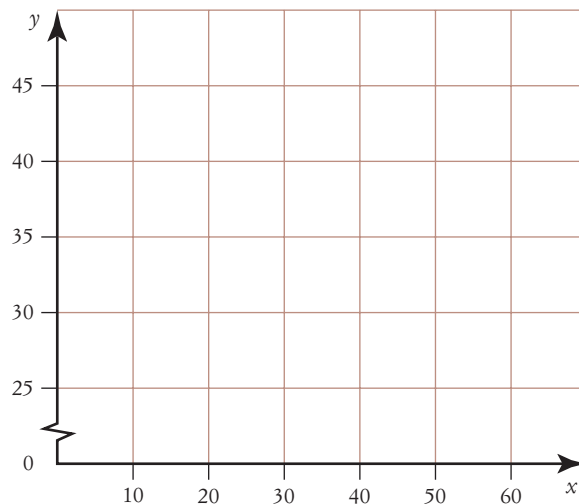
## Worksheet

# Scatter plots

1 Draw a scatter plot for each of the following sets of data.

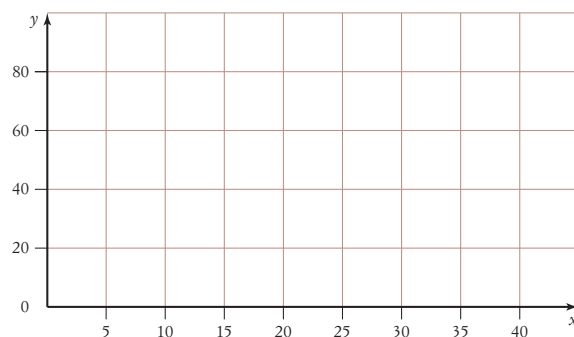
**a**

|     |    |    |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|----|----|
| $x$ | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 55 |
| $y$ | 45 | 39 | 38 | 36 | 34 | 37 | 31 | 29 | 32 |



**b**

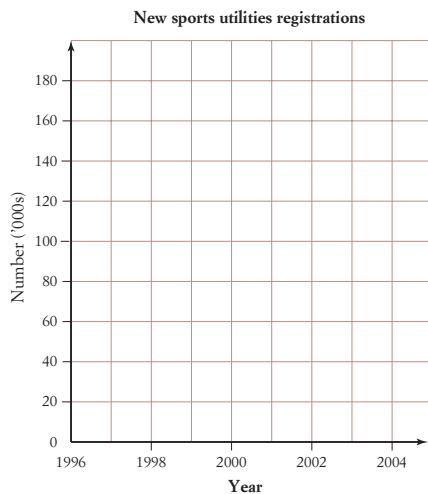
|     |    |    |    |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|----|----|----|
| $x$ | 3  | 7  | 11 | 15 | 19 | 23 | 27 | 31 | 35 | 39 |
| $y$ | 26 | 38 | 68 | 67 | 80 | 78 | 66 | 52 | 31 | 8  |



- 2 The following table shows the numbers of new sports utilities registered in Australia each year from 1997 to 2004.

| Year           | 1997 | 1998 | 1999 | 2000 | 2001 | 2002 | 2003 | 2004 |
|----------------|------|------|------|------|------|------|------|------|
| Number ('000s) | 72   | 96   | 104  | 106  | 118  | 139  | 150  | 173  |

- a Draw a scatter plot to show the results.



- b Does there seem to be any trend on your graph?

- c Draw in a line of best fit on the graph and use it to estimate the number of sports utilities that were registered in Australia in 1996. \_\_\_\_\_

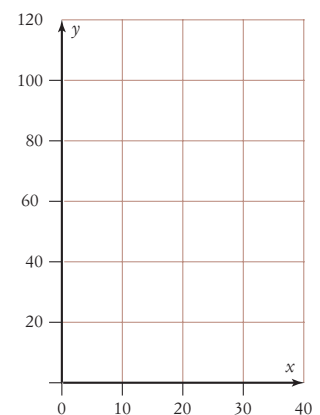
- 3 a Draw a scatter plot for each of the following sets of data.

- b State the nature of the relationship between the  $x$  values and  $y$  values for each data set.

i

| $x$ | 4 | 6  | 10 | 19 | 22 | 25 | 27 | 30 | 35  |
|-----|---|----|----|----|----|----|----|----|-----|
| $y$ | 9 | 21 | 28 | 63 | 67 | 79 | 84 | 84 | 104 |

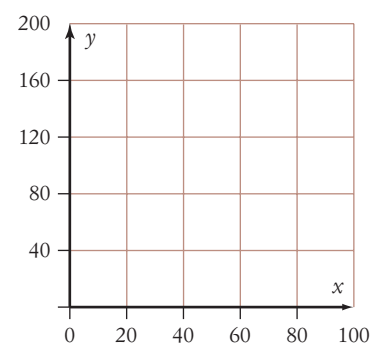
Relationship: \_\_\_\_\_



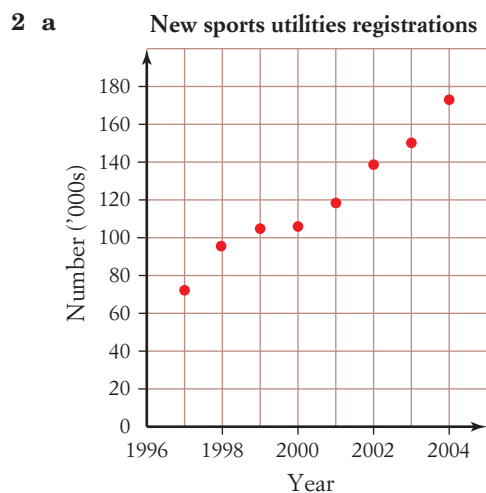
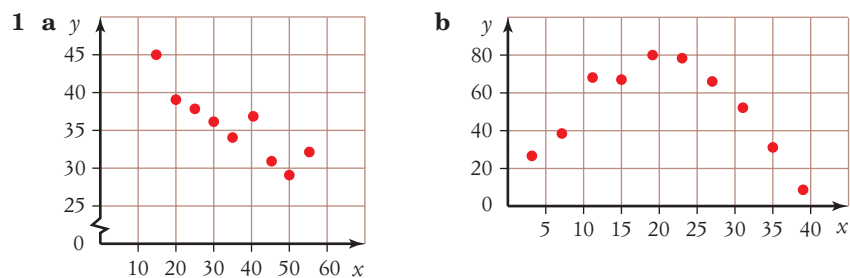
ii

| $x$ | 10  | 25  | 30  | 40  | 50 | 60 | 65 | 75 | 30  | 45  |
|-----|-----|-----|-----|-----|----|----|----|----|-----|-----|
| $y$ | 179 | 144 | 137 | 126 | 87 | 83 | 74 | 41 | 179 | 108 |

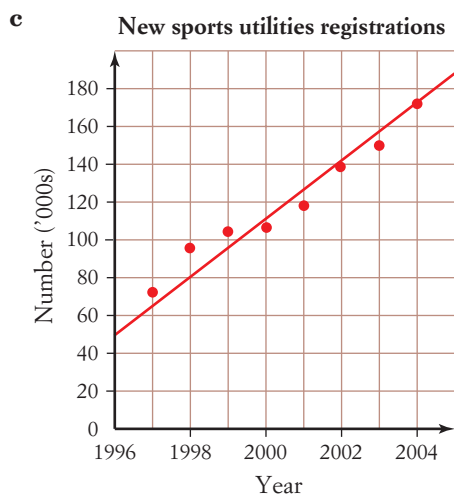
Relationship: \_\_\_\_\_



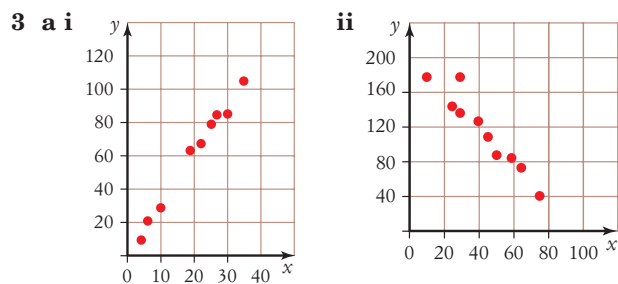
# Answers



**b** The number of sports utilities is increasing.



Approximately 50 000



**b i** Strong positive relationship      **ii** Strong negative relationship